Management of geological data and notifications in accordance with the Geological Data Act in Schleswig-Holstein

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Abstract: To cope with the tasks resulting from the Geological Data Act (GeolDG), the Schleswig-Holstein State Agency for the Environment (LfU) designed a Geological Notification and Management System (GAMS), which is intended to provide suitable support for the administration and processing of geological notifications and investigation data. The system presented, which is currently under development, is based on the data analysis and business intelligence platform Disy Cadenza Workbooks, into which a management application for controlling workflow processes, data checking and categorisation is integrated. Particular challenges lie in the implementation of numerous interfaces to components and external systems that are important for the editing process, such as the front-end web application for notifications of geological investigations. Dashboards have been designed in Cadenza Workbooks, which provide a convenient overview of essential data and enable the processing of notifications to be controlled.

Keywords: Geological Data Act, geological investigations, data management and data sharing, Cadenza Workbooks, dashboards

1 Introduction

The Geological Data Act (GeolDG, [Gz20], [RSH23]), which came into force in 2020, requires the State Agency for the Environment (LfU), as the responsible state authority for Schleswig-Holstein, to receive, review and publish notifications of geological investigations within specified deadlines. In addition, the geological data submitted following the investigations must be categorised according to their properties and their need for protection for public availability.

In terms of making data available, the Geological Data Act goes significantly beyond other existing legal regulations in environmental information, open data and access to

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information laws, as not only governmental data is affected, but the publication of data collected by the private sector is also prescribed: clients and contractors of geological investigations like drillings must report these before they are carried out and make the investigation data available afterwards [Gz20].

The aim is to permanently secure geological data and make it publicly available in order to ensure the sustainable management of the geological subsurface [Gz20].

In order to be able to guarantee legally compliant processing of the notifications and data deliveries within the tight time frame specified, continuous digital data processing and documentation is required from the notification of a geological investigation to the transfer of the associated technical and assessment data and their categorisation, combined with automated publication of the categorisation decision and the publication of the verification data.

This resulted in the need for the LfU to develop a management system that would provide suitable support for those responsible in the fulfilment of these tasks.

This article presents the Geological Notification and Management System (GAMS: Geologisches Anzeige- und Managementsystem), the development of which was initiated on the basis of the aforementioned task. Chapter 2 describes the requiremens and the architecture of the system followed by Chapter 3 dealing with the development steps and implementation. Chapter 4 concludes with a summary and an outlook.

2 Requirements and Architecture

2.1 Requirements for the geological notification management system

According to the Geological Data Act, notifications must be submitted at least 14 days before the start of a geological investigation. The relevant information on the planned investigation is transmitted with a notification, so that the management of the notifications essentially comprises the import of the notifications with structured, analysable storage of the information (*verification data*), support in checking the notifications and monitoring the subsequent transmission deadlines.

Once the geological investigations have been completed, both *technical data* and *assessment data* must be submitted to the LfU within certain deadlines. This leads to requirements for the management of these investigation results for the administration of this data and for support in checking the data. This includes in particular the *categorisation of the data*, which is to be carried out by the LfU and supported by GAMS in a suitable manner: All transmitted data must be assigned to categories in accordance with the Geological Data Act. The categorisation must be documented in an administrative act and made available to the public as well as to the notifying parties in a timely manner.

The overarching requirements therefore include monitoring the relevant deadlines, but also the provision of a message portal to enable communication with the involved parties and an interface to the electronic records system for audit-proof filing of the categorisation.

The areas of responsibility must be adequately mapped via a user and rights management system in order to be able to guarantee all data protection and data security requirements.

2.2 Overview of the basic components

The system was to be located within the central operating infrastructure of the Ministry of the Environment called ZeBIS (**Ze**ntrale **B**etriebs**i**nfra**s**truktur, see also [Ho21b]) in the TDC (Twin Data Centre) data centre of the state IT service provider Dataport[5] and also build on the standard components provided there as far as possible. These include central PostgreSQL database servers with PostGIS, on which the GAMS database is set up to manage all data, including geodata, and the ETL tool FME (Feature Manipulation Engine[6]) for transferring data and files between the platforms involved (see Figure 1). The use of certain specialised applications also requires the operation of an Oracle RDBMS within the LfU, so that data from the GAMS database must be transferred to Oracle as well as files must be transferred to the LfU using FME ETL processes.

Other ZeBIS components include a data warehouse database (DWH) for public data, which serves as a data source for OGC geodata services that are realised with the GDI software deegree[7].

The core management system consists of a PHP management application that is integrated into the Disy Cadenza Workbooks platform. Both components are explained in detail in the following sections (see sections 3.1 and 3.2).

Within the TDC data centre, further basic services such as secure file transfer, file service, mail server and Keycloak[8] are provided. Keycloak serves as a central identity and access management tool with a connection to the Active Directory (AD) of the state authorities offering single sign-on.

Outside the TDC, the Schleswig-Holstein Environmental Portal[9] is operated, which on the one hand forms the meta-information system for all environmental data, but on the other hand also makes all geodata accessible to the public via a WebGIS on the basis of the OGC services offered by deegree, including the drilling data [LfU25].

[5] Dataport AöR: https://www.dataport.de/who-we-are/, TDC: https://www.dataport.de/services-produkte/rechenzentrum/ accessed: 30/07/2025
[6] Safe Software: https://fme.safe.com/, accessed: 30/07/2025
[7] deegree: https://www.deegree.org/, accessed: 30/07/2025
[8] Keycloak: https://www.keycloak.org/, accessed: 30/07/2025
[9] Schleswig-Holstein Umweltportal: https://umweltportal.schleswig-holstein.de/, accessed: 30/07/2025

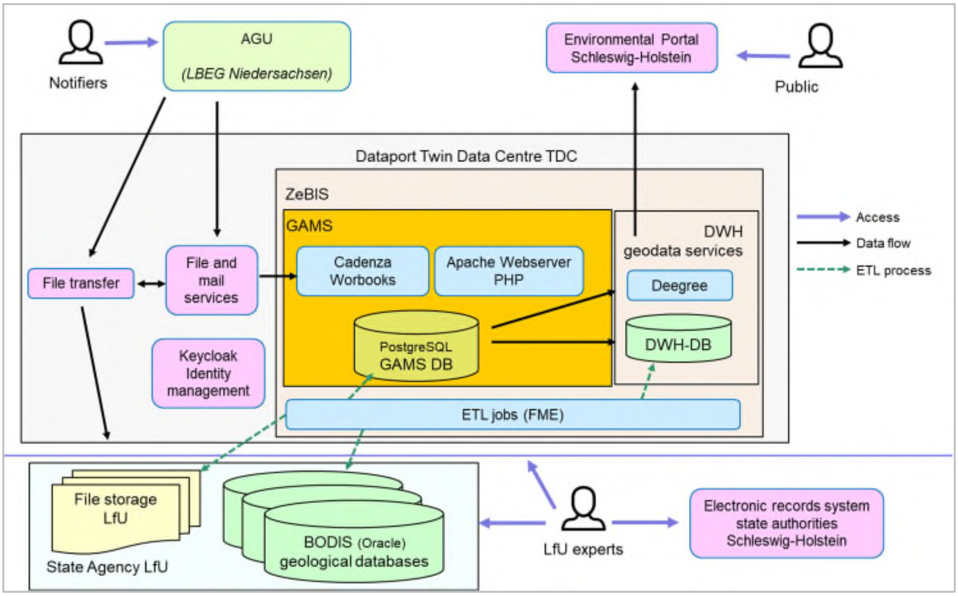

Figure 1:Architectural sketch of the most important components used for GAMS

3 Development steps and implementation

3.1 Navigation, analyses and WebGIS with Disy Cadenza Workbooks

The Disy Cadenza platform [DI25] is already being used for various tasks and issues in the Schleswig-Holstein environmental department. This includes specialised information systems based on Cadenza as well as map and factual data-based information and evaluation systems in the ZeBIS environment.

Disy Cadenza is a software which has been developed by the company Disy for many years on behalf of a co-operation of numerous federal states and other partners and adapted to current requirements and technical possibilities. It combines flexible evaluations, reporting and GIS functionalities in a software platform that is optimised for environmental data in particular [DI25].

In contrast to the previous - still widely used - Cadenza Classic, the current *Cadenza Workbooks* platform relies consistently on web technology with database-based configuration via a web interface. Supplementary web applications can be integrated directly into the Cadenza user interface in order to extend the evaluation platform in particular for editing data.

3.2 Integrated PHP management application to control the processes

While Cadenza Workbooks can be seen as a comprehensive construction kit for the flexible compilation of analyses and GIS functions, GAMS also requires a component that can implement more complex input functionalities as well as process controls developed individually for the management of notifications and geological data. This management application is implemented using the PHP[10] scripting language, which also allows flexible adaptations to technical requirements to be realised at short notice.

The combination of Cadenza with an integrated PHP specialised application to supplement application-specific functions has already proven to be a well suited solution in the Schleswig-Holstein environmental department in areas such as geology and soil, e.g. with the development of a consulting database geology and soil [Ho17] and a geotope register [Ho21a] as well as with other web applications for other departments of the LfU.

The PHP management application for GAMS is implemented on the basis of a framework developed by DigSyLand, which provides basic functionalities with regard to data security, catalogue processing, input validation and configurable data entry masks.

GAMS is particularly innovative because the management application is one of the first specialised applications to be integrated directly into Cadenza Workbooks via the Web API Workbooks, as the application framework and user management of Cadenza Classic were previously always used. Therefore, significant new developments had to be created with the connection via the Web API Workbooks and with the OpenID Connect interface to Keycloak as the central identity provider for GAMS. On the basis of this connection, which realises a single sign-on (SSO) of Cadenza Workbooks and the management application, also direct linkage between dashboards and input masks is possible.

3.3 Interface to the notification web application AGU

The submission of the notifications and data of the geological investigations is performed by the notifiers using the web application AGU[11] (*Anzeige Geologischer Untersuchungen*: notification of geological investigations) operated by the State Agency for Mining, Energy and Geology (LBEG[12]) of Lower Saxony (s. Figure 1). The LBEG provides this front-end application for the states of Lower Saxony, Bremen, Hamburg, and Schleswig-Holstein as well as for the Federal Institute for Geosciences and Natural Resources (BGR). Submitted notifications including all data files are sent by the AGU to the responsible state authorities.

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[10] PHP: https://www.php.net/, accessed: 30/07/2025

[11] Anzeige Geologischer Untersuchungen (AGU): https://nibis.lbeg.de/agu/startseite, accessed: 30/07/2025

[12] Landesamt für Bergbau, Energie und Geologie (LBEG) des Landes Niedersachsen: https://www.lbeg.niedersachsen.de/, accessed: 30/07/2025

It is planned that the GAMS application will receive the notifications by the AGU by e-mail or file transfer as XML files. GAMS carries out checks for formal consistency and stores the notification data in the PostgreSQL GAMS database (see Figure 1). This also includes the transfer of the transmitted geometries of the investigation area, so that a spatial analysis by the specialists is possible with GAMS (e.g. boreholes in so-called risk areas). Once the data has been imported, subsequent processes such as e-mail notifications to the responsible employees at the LfU and the notifying parties are automatically triggered via the message portal implemented in the GAMS management PHP application. The basic routines of these processes have already been developed, mainly used for a predecessor application of AGU, and are now being adapted to the current AGU and GAMS applications.

A secure file transfer from the LBEG to the GAMS data centre is planned for the transfer of additional files with the investigation data, so that the transferred files can be analysed using the PHP management application, which supports the examination by the experts. Finally, the files with the examination data are transferred to the internal LfU network for storage using an ETL process (see Figure 1).

3.4 Evaluation dashboards

Comprehensive, customisable evaluations of the data are required in order to optimally carry out the technical assessment of the reports and investigation data. This includes map visualisations as well as the selection, analysis and further processing of notification data sets selected on the map. The evaluation criteria and the compilation of the result presentations as tables, maps and diagrams as well as reports should be customisable by the specialist managers themselves as required.

Therefore dashboards are implemented in Cadenza Workbooks offering the essential information in a suitable display form at a glance, so that, for example, depending on the processing deadlines, the data records to be processed as a matter of priority are highlighted with the option of calling them up directly in the GAMS management application for processing. Lists of the queried records are added by map views and diagrams.

3.5 Further current development steps

The input and editing masks of the GAMS management application are currently being completed, in particular with functionalities for realising the notication management workflows. This includes automatic checking, e. g. for risk areas based on geometries, unrealistic depths, etc. as well as support for the manual processing steps for formal checking, confirmation and publication, including the option of requesting additional information from the notifier. It will provide suitable support for the legally prescribed classification of the investigation data and the determination of the data category by offering processing masks and checklists.

In addition to maps of the location of geological investigations the information provided on the Internet also includes the categorisation notices, which are generated from the management application as accessible (barrier-free) PDF files. The categorisation notice documents the submitted and checked verification data as well as the list of the submitted data with categorisation as *verification data*, *technical data* or *assessment data* including further classifications like reasons for protection. As the categorisation notices are made publicly available as part of the official administrative act, it was an essential requirement to create the corresponding PDF files in an accessible format. It is also planned to store these files belonging to the corresponding official administrative act in the state's e-file (electronic records system) in an audit-proof manner (see Figure 1).

4 Summary and outlook

The GAMS Geological Notification and Management System is designed to efficiently support the State Agency for the Environment in Schleswig-Holstein as the responsible authority in the management of notifications and geological investigation data.

The automatic import into the GAMS database with structuring of the supplied metadata and basic consistency checks facilitates the technical review for those responsible, who are also supported by customised evaluation options in Cadenza Workbooks.

Centrally curated dashboards with tables, diagrams and maps as well as views and evaluations customised by the users themselves enable flexible responses to current requirements. The GAMS database thus provides a standardised database that is required to fulfil the obligations of the Geological Data Act by the state agency.

The IT components in the central operational infrastructure ZeBIS provided for by the framework concept of the Ministry of the Environment can be suitably combined in GAMS, so that GAMS can be considered a good example of the application of the framework concept.

Main challenges include the linkage with internal and external systems and components providing a number of interfaces, such as for the ID provider Keycloak and the AGU web frontend.

The first installation on the test platform in the ZeBIS environment in the data center, with connection to the basic components, is planned for mid-2025. Going live is scheduled for the end of 2025.

Bibliography

[DI25] Disy Informationssysteme GmbH: Software for Data Analytics, Business & Location Intelligence, https://www.disy.net/en/products/disy-cadenza/overview/, accessed: 30/07/2025.

[Gz20] Gesetz zur staatlichen geologischen Landesaufnahme sowie zur Übermittlung, Sicherung und öffentlichen Bereitstellung geologischer Daten und zur Zurverfügungstellung geologischer Daten zur Erfüllung öffentlicher Aufgaben (Geologiedatengesetz - GeolDG) - Geologiedatengesetz vom 19. Juni 2020 (BGBl. I S. 1387), 2020, https://www.gesetze-im-internet.de/geoldg/, accessed: 30/07/2025

[Ho17] Hosenfeld, F. et al.: Consulting Database Geology and Soil in Schleswig-Holstein. In (Otjacques, B. et al., eds.): From Science to Society: The Bridge provided by Environmental Informatics, Adjunct Proceedings of the 31st EnviroInfo conference, Luxembourg, pp. 173-179, 2017.

[Ho21a] Hosenfeld, F. et al.: Neuentwicklung einer Intranet-Web-Anwendung für das Geotopkataster Schleswig-Holstein. In (Freitag, U. et al., eds.): Umweltinformations-systeme - Wie verändert die Digitalisierung unsere Gesellschaft? Tagungsband des 27. Workshops „Umweltinformationssysteme (UIS2020)“ des Arbeitskreises „Umweltinformationssysteme“ der Fachgruppe „Informatik im Umweltschutz‘‘ der Gesellschaft für Informatik (GI), Springer Vieweg, Wiesbaden, pp. 13-27, 2021, https://doi.org/10.1007/978-3-658-30889-6_2, accessed: 30/07/2025.

[Ho21b] Hosenfeld, F. et al.: Governmental Information Systems Supporting Nature Conservation in Schleswig-Holstein. In (Wohlgemuth, V. et al., eds.): Environmental Informatics. A bogeyman or saviour to achieve the UN Sustainable Development Goals? Adjunct Proceedings of the 35th edition of the EnviroInfo – the long standing and established international and interdisciplinary conference series on leading environmental information and communication technologies Berlin, Germany, September 27-29, pp. 133-140, 2021.

[LfU25] Landesamt für Umwelt Schleswig-Holstein: Bohrungen in Schleswig-Holstein. Metadatensatz im Umweltportal Schleswig-Holstein, https://umweltportal.schleswig-holstein.de/trefferanzeige?docuuid=ffa99fdc-8548-41fd-b10a-a165ff905aee, accessed: 30/07/2025.

[RSH23] Reiche, S.; Schröder, L. A.; Hermann, D.: The collection, digitisation, interpretation and publication of geological data in the German site selection procedure – status and challenges. In (Dietl, C. et al., eds.): Interdisciplinary Research Symposium on the Safety of Nuclear Disposal Practices safeND 2023, Berlin, 13–15 September 2023, Safety of Nuclear Waste Disposal, Volume 2, pp. 141–141, 2023, https://doi.org/10.5194/sand-2-141-2023, accessed: 30/07/2025.